Introduction to NLP, Exercise 2

Sabine Weber

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Discussion: Nov 5, 2026

- **Deadline:** The submission needs to be done via VC until the end of the day mentioned on the PDF. A late submission is not possible.
- **Groups:** You are allowed to work in groups. A group can consist of maximally three members. The members can change in each submission. The names that you put on the PDF are what we consider as the group.
- **Grading:** We plan to offer 11 exercise sheets. For each sheet, you can achieve 20 points. From the 220 points altogether, we will consider 176 points to be a perfect participation (if, for some reason, we offer fewer assignments, this number of required points will be reduced proportionally). The assignment submissions will contribute up to 20 % of the grade as bonus points.
- **Submission:** Only submit one PDF file. Program code should be part of that PDF file. Put all names and email addresses of your group members on the first page of this PDF file. Please do submit the PDF only once (not multiple times by multiple group members).
- **Sources:** We recommend to not use any "AI" or large language model for creating the solution to any subtask/task as a whole. You can use other sources (documentation, papers, stackexchange, ...). If you decide to use an "AI" or LLM (like ChatGPT, Hawki) for a part of your solution, provide the information which system you used, provide the complete prompt and the complete answer by the automatic system (in German or English) in an appendix of your submission. I need to be able to understand what your own contribution on top of the automatic result was. You only receive points for your own contribution on top of what you got from the source you provide. If you use an "AI" and do not provide the required reference or do not make that use transparent, we will consider this plagiarism.
- **Plagiarism:** When two groups submit solutions that are copied from each other or from a third source that is not made transparent in the submission, all these groups do not receive any points. The same holds if one group copies from a source that is no provided in the submission. The whole submission will be considered invalid, not only the task in which we detected the plagiarism. No points will be given for the whole submission. These sanctions do not exclude further measures.

Do you have questions? Send me an email (sabine.weber@uni-bamberg.de) or use the forum.

Task 1 - Pen & Paper (10 points)

Implement the Byte Pair Encoding algorithm manually for a small corpus.

Consider the following corpus:

Word	Frequency
low	5
lower	2
lowest	3
newer	3
wider	2
widest	3

- Represent each word as a sequence of characters.
- Treat each word as a separate sequence.
- Use word frequencies as weights.
- Do not count pairs across word boundaries.
- Perform six BPE merge operations.
- If several pairs have the same highest frequency, select the pair whose concatenated string comes first alphabetically.

Tokenize unseen word `lowers`. What is its final tokenization? Reflect: Does the learned tokenization reflect the morphological structure of the words? Is tokenization into morphemes a desirable property for downstream NLP tasks, and why or why not?

Task 2 - Programming (10 points)

Implement and compare two different approaches to word segmentation.

a) Rule-based stemming

- Design your own stemming rules based on common prefixes and suffixes. Do not use an existing stemming library.
- Test your rules on at least 10 words, including inflected forms and words with similar endings but different meanings.

b) Byte Pair Encoding (BPE)

- Implement the BPE algorithm for word segmentation.
- Select two different corpora to learn byte pair encodings from, one in English and one in German. Create a separate BPE model for each corpus.

Next, create a test set of German and English sentences and apply the rule-based stemming rules as well as both BPE segmentations (German and English) to the sentences. What happens when the language of the sentence does not match the language of the BPE model? Compare the results and discuss strengths and weaknesses of each approach.